

Interview Prep Platform — Use Case Analysis

AI-Powered Interview Practice with Intelligent Scoring & Feedback

Actors & Use Cases

Primary Actors

Job Seeker (Primary — initiates core workflows)

CS students, bootcamp graduates, early-career engineers, career switchers.

Use cases:

Select job role Choose SWE, Data Science, PM, or Behavioral track

Practice interview Receive curated questions for the selected role

Submit answer Type a text response to each interview question

View score & feedback See 4-axis score: Technical Depth, Clarity, Completeness, Structure

Track progress Review answer history, radar charts, and score trends over time

Secondary Actors

University / Career Center (Secondary — institutional access)

Career centers and CS departments offering guided interview prep resources.

Use cases:

Manage question bank Access or curate role-specific question sets for students

AI / ML System (Secondary — internal scoring engine)

sentence-transformers model + cosine similarity scorer + keyword extractor.

Use cases:

Authenticate user JWT-based session management via FastAPI auth layer

Score answer (AI) Embed answer & rubric; compute cosine similarity score

Generate feedback Surface missing rubric elements as actionable improvement tips

Key Relationships

Submit answer → **Score answer (AI)**

«includes» Submitting always triggers AI scoring — mandatory sub-flow.

Score answer (AI) → **Generate feedback**

«includes» Scoring always triggers feedback generation — mandatory sub-flow.

Interview Prep Platform — Use Case Diagram

High-Level System Boundary · Actors · Use Cases · Relationships

